

# Magnetic Fields — Paper A

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Write in sentences — a list of single words scores badly on this topic.
- Every field line you draw must carry an arrowhead. A line without one does not score.
- Where a question asks for a number, that number must carry a unit.
- The mark for each question is shown in square brackets on the right.
- Reminder: outside a magnet the field lines run from north to south, like poles repel and unlike poles attract.

## SECTION 1 – MATERIALS AND POLES

**1.** Name two magnetic materials, and two metals that are not magnetic. [2]

**2.** Complete the table. Write yes or no in each empty box. [3]

| Material  | Attracted by a magnet? |
|-----------|------------------------|
| iron      |                        |
| copper    |                        |
| nickel    |                        |
| aluminium |                        |
| cobalt    |                        |
| brass     |                        |

**3.** State what happens in each case. [1]  
**(a)** The north poles of two magnets are brought together.

**(b)** The south poles of two magnets are brought together. [1]

**(c)** The north pole of one magnet is brought near the south pole of another. [1]

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**4.** Explain the difference between a magnetic material and a magnet. [2]

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SECTION 2 – THE MAGNETIC FIELD

**5.** A bar magnet lies on the page with its north pole on the left.

**(a)** Draw the magnetic field pattern around it. Draw at least four lines. [3]

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**(b)** State which pole your arrows point away from. [1]

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**6.** State three rules that must be obeyed when drawing magnetic field lines. [3]

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**7.** On a correct diagram the field lines are drawn closer together at the poles than at the sides. State what this tells you, and why the spacing is worth marks. [2]

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- 8.** Describe how you would use a plotting compass to plot one magnetic field line around a bar magnet. [3]

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SECTION 3 – INDUCED MAGNETISM AND THE EARTH

- 9.** Explain what is meant by induced magnetism. [2]

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- 10.** State one difference between soft iron and steel, and name one thing each is used to make. [2]

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- 11.** Explain why the needle of a compass always comes to rest pointing north. [2]

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- 12.** State the test that proves an object is a magnet, and explain why attraction on its own is not enough. [2]

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*End of paper. Before you hand it in, go back over every diagram and check that each field line carries an arrowhead pointing from north to south.*